

Merrivoss Bioanalysis Inc, a Tarnovell company

# TP04 inflammatory kinase follow-up: 40-compound cardiac and CYP screen

## Final Report

Study MV-2024-358 (LIMS M24/BIO/08257)

Client: Tervemere Pharma AG

Study lead: Dr P. Nandekuri, DMPK Lead

Report date: December 19, 2024

## 1 Summary

40 compounds from the Tervemere aminopyrazolopyrimidine series 04 were tested for Tervemere Pharma AG in the IKr inhibition, Five-isoform DDI screen, CYP3A4 inhibition curves. Testing started on December 5, 2024 and results were reported on December 19, 2024. All accepted runs met matrix acceptance criteria. No compound had a hERG IC50 below 1 µM. Most compounds (31 of 40) had IC50 at or above 10 µM or above the top concentration. Inhibition above 50% at 10 µM was most frequent for CYP3A4.

Table 1 Study summary

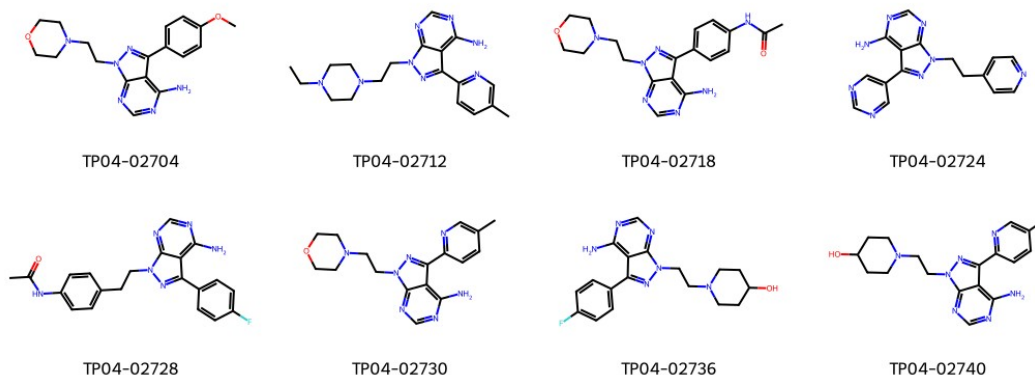
| Metric                         | Value                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------|
| Project                        | MV-2024-358                                                                                    |
| Client                         | Tervemere Pharma AG                                                                            |
| Compounds tested               | 40                                                                                             |
| Series / chemotype             | Tervemere aminopyrazolopyrimidine series 04 (Aminopyrazolopyrimidine)                          |
| Assays                         | hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM, CYP3A4 inhibition IC50 |
| Turnaround (calendar days)     | 14                                                                                             |
| Price                          | USD 24,700                                                                                     |
| hERG IC50 <1 µM                | 0                                                                                              |
| hERG IC50 1-10 µM              | 9                                                                                              |
| hERG IC50 ≥10 µM               | 31                                                                                             |
| hERG median IC50 (µM)          | 16                                                                                             |
| E-4031 control IC50 (nM)       | 15                                                                                             |
| CYP1A2 >50% inhibition @10 µM  | 5                                                                                              |
| CYP2C9 >50% inhibition @10 µM  | 6                                                                                              |
| CYP2C19 >50% inhibition @10 µM | 7                                                                                              |
| CYP2D6 >50% inhibition @10 µM  | 5                                                                                              |
| CYP3A4 >50% inhibition @10 µM  | 9                                                                                              |
| Compounds with any CYP >50%    | 12                                                                                             |
| Ketoconazole (3A4) IC50 (µM)   | 0.022                                                                                          |

## 2 Objective

To rank the test article set for IKr inhibition, Five-isoform DDI screen, CYP3A4 inhibition curves liability so that Tervemere Pharma AG could choose analogues for progression.

## 3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on December 5, 2024 and logged against M24/BIO/08257. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



## 4 Methods

### IKr inhibition

CHO cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarized to +40 mV for 2 s and repolarized to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30  $\mu$ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Matrix acceptance: seal resistance  $\geq 500$  M $\Omega$  before break-in and  $\geq 200$  M $\Omega$  throughout
- Tail current  $\geq 200$  pA at baseline; run-down  $\leq 10\%$  over the vehicle period
- E-4031 IC<sub>50</sub> within 5-50 nM; runs outside this range are rejected and repeated
- IC<sub>50</sub> fitted per cell (four-parameter logistic); the compound IC<sub>50</sub> is the geometric mean of finite cell IC<sub>50</sub>s when these are the majority of accepted cells, otherwise reported as  $>30$   $\mu$ M
- Where precipitation is seen at 30  $\mu$ M, that concentration is excluded and the highest evaluable concentration is reported

## Five-isoform DDI screen

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10  $\mu$ M and an isoform-selective probe substrate at approximately  $K_m$ , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

### Probe substrates and controls

| Isoform | Probe substrate  | Metabolite            | Positive control |
|---------|------------------|-----------------------|------------------|
| CYP1A2  | Phenacetin       | Acetaminophen         | Furafylline      |
| CYP2C9  | Diclofenac       | 4'-Hydroxydiclofenac  | Sulfaphenazole   |
| CYP2C19 | S-Mephenytoin    | 4'-Hydroxymephenytoin | Ticlopidine      |
| CYP2D6  | Dextromethorphan | Dextrorphan           | Quinidine        |
| CYP3A4  | Midazolam        | 1'-Hydroxymidazolam   | Ketoconazole     |

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC<sub>50</sub> falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10  $\mu$ M are flagged, and small negative values indicate no inhibition.

## CYP3A4 inhibition curves

All compounds were run in a 7-point CYP3A4 concentration response (top concentration 50  $\mu$ M, 3-fold dilutions) using midazolam 1'-hydroxylation as the probe, with ketoconazole on every plate. A single probe is used so that IC<sub>50</sub>s are comparable across the set; substrate-dependent differences are outside the scope.

## 5 Results

### hERG results

All 40 compounds were tested. No compound had a hERG IC<sub>50</sub> below 1  $\mu$ M. Most compounds (31 of 40) had IC<sub>50</sub> at or above 10  $\mu$ M or above the top concentration. The median IC<sub>50</sub> was 16  $\mu$ M (0 below 1  $\mu$ M, 9 between 1 and 10  $\mu$ M, 31 at or above 10  $\mu$ M, of which 11 above the highest evaluable concentration). E-4031 gave an IC<sub>50</sub> of 15 nM in the accepted runs.

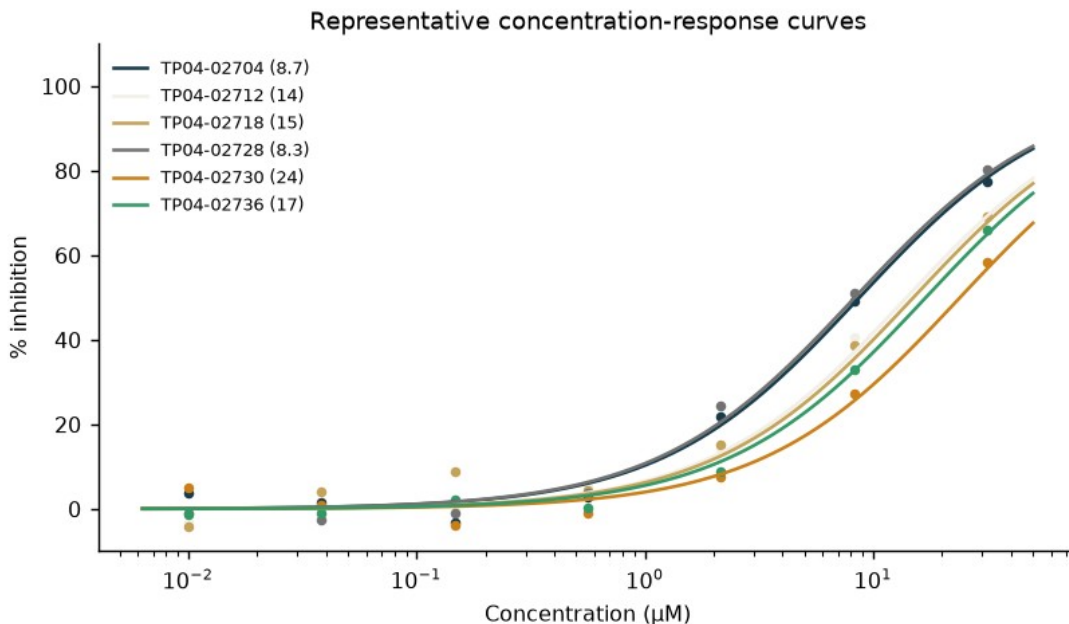
**Table 2 hERG IC<sub>50</sub> by compound ( $\mu$ M)**

| Compound   | IC <sub>50</sub> cell 1 ( $\mu$ M) | IC <sub>50</sub> cell 2 ( $\mu$ M) | IC <sub>50</sub> cell 3 ( $\mu$ M) | IC <sub>50</sub> geomean ( $\mu$ M) | % inhibition @10 $\mu$ M | n cells (finite) | QC note |
|------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|--------------------------|------------------|---------|
| TP04-02704 | 8.7                                | 6.6                                |                                    | 7.6                                 | 59                       | 2 (2)            |         |
| TP04-02712 | 14                                 | 9.8                                |                                    | 12                                  | 40                       | 2 (2)            |         |
| TP04-02718 | 15                                 | 14                                 |                                    | 14                                  | 40                       | 2 (2)            |         |
| TP04-02724 | >30                                | >30                                |                                    | >30                                 | 25                       | 2 (0)            |         |
| TP04-02728 | 8.3                                | 7.2                                |                                    | 7.7                                 | 59                       | 2 (2)            |         |
| TP04-02730 | 24                                 | 28                                 |                                    | 26                                  | 32                       | 2 (2)            |         |
| TP04-02736 | 17                                 | 13                                 | 16                                 | 15                                  | 42                       | 3 (3)            |         |
| TP04-02740 | 15                                 | 15                                 | 9.8                                | 13                                  | 35                       | 3 (3)            |         |

|            |     |     |     |     |    |       |  |
|------------|-----|-----|-----|-----|----|-------|--|
| TP04-02748 | 16  | 16  |     | 16  | 41 | 2 (2) |  |
| TP04-02752 | 14  | 12  |     | 13  | 35 | 2 (2) |  |
| TP04-02764 | 10  | 5.7 |     | 7.7 | 53 | 2 (2) |  |
| TP04-02766 | 11  | 16  | 28  | 17  | 39 | 3 (3) |  |
| TP04-02770 | >30 | >30 |     | >30 | 27 | 2 (0) |  |
| TP04-02772 | 5.4 | 4.4 |     | 4.9 | 67 | 2 (2) |  |
| TP04-02776 | 24  | 21  | >30 | 23  | 33 | 3 (2) |  |
| TP04-02782 | 29  | >30 |     | >30 | 18 | 2 (1) |  |
| TP04-02784 | 6.8 | 14  | 13  | 11  | 47 | 3 (3) |  |
| TP04-02794 | 7.9 | 6.7 |     | 7.3 | 55 | 2 (2) |  |
| TP04-02800 | 17  | >30 |     | >30 | 27 | 2 (1) |  |
| TP04-02806 | >30 | >30 |     | >30 | 20 | 2 (0) |  |
| TP04-02812 | 18  | 10  |     | 14  | 40 | 2 (2) |  |
| TP04-02814 | 16  | 16  |     | 16  | 37 | 2 (2) |  |
| TP04-02820 | 26  | 22  |     | 24  | 32 | 2 (2) |  |
| TP04-02824 | 7.6 | 4.2 |     | 5.6 | 66 | 2 (2) |  |
| TP04-02826 | >30 | >30 |     | >30 | 29 | 2 (0) |  |
| TP04-02830 | 6.3 | 6.4 |     | 6.4 | 56 | 2 (2) |  |
| TP04-02832 | 19  | 20  |     | 19  | 41 | 2 (2) |  |
| TP04-02842 | 22  | 29  |     | 25  | 28 | 2 (2) |  |
| TP04-02844 | 11  | 8.1 |     | 9.5 | 54 | 2 (2) |  |
| TP04-02848 | 6.7 | 7.7 |     | 7.1 | 54 | 2 (2) |  |
| TP04-02850 | 16  | 19  |     | 17  | 40 | 2 (2) |  |
| TP04-02856 | 14  | 13  |     | 14  | 43 | 2 (2) |  |
| TP04-02860 | 11  | 9.3 |     | 10  | 50 | 2 (2) |  |
| TP04-02862 | >30 | >30 | 26  | >30 | 31 | 3 (1) |  |
| TP04-02866 | >30 | >30 |     | >30 | 25 | 2 (0) |  |
| TP04-02868 | 28  | 22  |     | 25  | 30 | 2 (2) |  |
| TP04-02878 | 23  | 25  | 18  | 22  | 29 | 3 (3) |  |
| TP04-02884 | >30 | >30 |     | >30 | 21 | 2 (0) |  |
| TP04-02886 | >30 | >30 |     | >30 | 15 | 2 (0) |  |
| TP04-02890 | 21  | >30 |     | >30 | 30 | 2 (1) |  |

Top concentration 30  $\mu$ M; 6-point, 3-fold dilution; positive control E-4031 IC50 15 nM (acceptance 5-50 nM)

Compound IC50 = geometric mean of finite cell IC50s when they are the majority of accepted cells; otherwise reported as >30  $\mu$ M (or >10  $\mu$ M where solubility-limited)



### CYP inhibition results

Inhibition above 50% at 10  $\mu\text{M}$  was most frequent for CYP3A4. In total 12 compounds inhibited at least one isoform by more than 50% at 10  $\mu\text{M}$  (CYP3A4: 9; CYP2D6: 5; CYP2C19: 7; CYP2C9: 6; CYP1A2: 5). All control IC<sub>50</sub>s were within 0.5-2 times the historical mean (ketoconazole 0.022  $\mu\text{M}$  against a historical mean of 0.030  $\mu\text{M}$ ).

**Table 3 Percent inhibition at 10  $\mu\text{M}$**

| Compound   | CYP1A2 (%) | CYP2C9 (%) | CYP2C19 (%) | CYP2D6 (%) | CYP3A4 (%) |
|------------|------------|------------|-------------|------------|------------|
| TP04-02704 | 25         | 0          | 8           | 2          | 21         |
| TP04-02712 | 1          | 12         | 5           | 3          | 3          |
| TP04-02718 | 3          | 7          | 0           | 2          | 4          |
| TP04-02724 | 42         | 6          | 8           | 51         | 47         |
| TP04-02728 | 37         | 91         | 100         | 63         | 85         |
| TP04-02730 | -1         | -3         | 9           | 4          | -2         |
| TP04-02736 | 4          | 3          | 4           | 2          | 9          |
| TP04-02740 | -6         | 3          | -5          | 15         | 6          |
| TP04-02748 | 2          | 1          | 11          | 1          | 3          |
| TP04-02752 | 17         | 86         | 39          | 5          | 56         |
| TP04-02764 | 57         | 21         | 58          | 95         | 81         |
| TP04-02766 | 9          | 11         | 3           | -8         | 11         |
| TP04-02770 | -2         | 7          | -1          | 1          | 28         |
| TP04-02772 | 72         | 17         | 11          | 4          | 6          |
| TP04-02776 | 21         | 1          | 13          | 4          | -2         |
| TP04-02782 | -5         | -1         | 0           | 10         | 40         |
| TP04-02784 | -8         | 6          | -1          | -0         | -6         |
| TP04-02794 | 7          | 8          | 6           | 8          | 4          |
| TP04-02800 | 90         | 55         | 88          | 100        | 97         |
| TP04-02806 | 10         | 4          | -1          | 1          | -0         |
| TP04-02812 | 3          | 17         | -0          | -11        | -0         |
| TP04-02814 | 6          | 2          | -2          | -0         | 9          |
| TP04-02820 | 6          | 3          | -0          | 5          | 22         |
| TP04-02824 | 6          | 7          | -2          | -6         | 9          |
| TP04-02826 | 4          | 11         | -0          | 3          | -2         |

|            |    |    |    |     |    |
|------------|----|----|----|-----|----|
| TP04-02830 | 21 | 95 | 77 | 24  | 91 |
| TP04-02832 | 31 | 97 | 83 | 30  | 98 |
| TP04-02842 | 85 | 37 | 65 | 90  | 81 |
| TP04-02844 | 5  | 16 | 2  | 6   | -2 |
| TP04-02848 | 75 | 3  | 25 | 6   | 12 |
| TP04-02850 | -1 | 4  | 1  | 5   | 9  |
| TP04-02856 | 3  | 6  | -0 | -2  | -6 |
| TP04-02860 | 24 | 96 | 67 | 34  | 80 |
| TP04-02862 | -0 | -3 | 0  | -5  | 7  |
| TP04-02866 | 3  | -7 | 1  | -7  | 2  |
| TP04-02868 | 1  | 4  | 5  | 2   | 12 |
| TP04-02878 | 11 | 37 | 36 | 2   | 75 |
| TP04-02884 | 8  | 15 | 14 | 0   | 6  |
| TP04-02886 | -2 | 5  | 8  | -12 | -3 |
| TP04-02890 | 7  | -5 | 2  | 4   | 11 |

Single concentration 10  $\mu$ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC<sub>50</sub>,  $\mu$ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.38; ticlopidine (CYP2C19) 1.4; quinidine (CYP2D6) 0.041; ketoconazole (CYP3A4) 0.022

**Table 4 Positive control IC<sub>50</sub>s**

| Control        | Isoform | IC <sub>50</sub> ( $\mu$ M) | Historical mean ( $\mu$ M) |
|----------------|---------|-----------------------------|----------------------------|
| Furafylline    | CYP1A2  | 1.9                         | 1.8                        |
| Sulfaphenazole | CYP2C9  | 0.38                        | 0.35                       |
| Ticlopidine    | CYP2C19 | 1.4                         | 1.2                        |
| Quinidine      | CYP2D6  | 0.041                       | 0.060                      |
| Ketoconazole   | CYP3A4  | 0.022                       | 0.030                      |

### CYP3A4 concentration response

CYP3A4 IC<sub>50</sub>s were determined for all 40 compounds: 10 had IC<sub>50</sub> below 10  $\mu$ M, including 4 below 1  $\mu$ M. These potent inhibitors are the main CYP liability in the set.

**Table 5 CYP3A4 IC<sub>50</sub> ( $\mu$ M)**

| Compound   | IC <sub>50</sub> ( $\mu$ M) | Probe substrate |
|------------|-----------------------------|-----------------|
| TP04-02704 | 36                          | midazolam       |
| TP04-02712 | >50                         | midazolam       |
| TP04-02718 | >50                         | midazolam       |
| TP04-02724 | 8.1                         | midazolam       |
| TP04-02728 | 2.7                         | midazolam       |
| TP04-02730 | >50                         | midazolam       |
| TP04-02736 | >50                         | midazolam       |
| TP04-02740 | >50                         | midazolam       |
| TP04-02748 | >50                         | midazolam       |
| TP04-02752 | 5.2                         | midazolam       |
| TP04-02764 | 3.9                         | midazolam       |
| TP04-02766 | >50                         | midazolam       |
| TP04-02770 | 22                          | midazolam       |
| TP04-02772 | >50                         | midazolam       |
| TP04-02776 | >50                         | midazolam       |
| TP04-02782 | 18                          | midazolam       |
| TP04-02784 | >50                         | midazolam       |
| TP04-02794 | >50                         | midazolam       |
| TP04-02800 | 0.20                        | midazolam       |

|            |      |           |
|------------|------|-----------|
| TP04-02806 | >50  | midazolam |
| TP04-02812 | >50  | midazolam |
| TP04-02814 | >50  | midazolam |
| TP04-02820 | 47   | midazolam |
| TP04-02824 | >50  | midazolam |
| TP04-02826 | >50  | midazolam |
| TP04-02830 | 0.78 | midazolam |
| TP04-02832 | 0.21 | midazolam |
| TP04-02842 | 0.94 | midazolam |
| TP04-02844 | >50  | midazolam |
| TP04-02848 | >50  | midazolam |
| TP04-02850 | >50  | midazolam |
| TP04-02856 | >50  | midazolam |
| TP04-02860 | 2.4  | midazolam |
| TP04-02862 | >50  | midazolam |
| TP04-02866 | >50  | midazolam |
| TP04-02868 | >50  | midazolam |
| TP04-02878 | 3.7  | midazolam |
| TP04-02884 | >50  | midazolam |
| TP04-02886 | >50  | midazolam |
| TP04-02890 | >50  | midazolam |

7-point curve, top 50  $\mu\text{M}$ ; ketoconazole control run alongside on every plate

## 6 Deviations

No deviations from the study plan.

## 7 Discussion

No compound had a hERG IC<sub>50</sub> below 1  $\mu\text{M}$ . Most compounds (31 of 40) had IC<sub>50</sub> at or above 10  $\mu\text{M}$  or above the top concentration. Inhibition above 50% at 10  $\mu\text{M}$  was most frequent for CYP3A4. Values are in vitro and do not by themselves predict clinical risk; free exposure estimates are needed for that assessment.

## 8 Conclusion

Recommended next steps: prioritize compounds with hERG IC<sub>50</sub> at or above 10  $\mu\text{M}$ ; for any compound below 10  $\mu\text{M}$ , compare the IC<sub>50</sub> with projected free plasma exposure before progression; deprioritize compounds with CYP3A4 IC<sub>50</sub> below 1  $\mu\text{M}$  and review the below-10  $\mu\text{M}$  group with the chemistry team.

## 9 Records

Raw data, assay worksheets and instrument files are retained for 5 years under M24/BIO/08257.

| Name            | Role      | Signature          | Date              |
|-----------------|-----------|--------------------|-------------------|
| Priya Nandekuri | DMPK Lead | <u>P Nandekuri</u> | December 19, 2024 |

|                 |                               |                    |                   |
|-----------------|-------------------------------|--------------------|-------------------|
| Helen Marchwood | Bioanalytical Quality Manager | <u>H Marchwood</u> | December 18, 2024 |
|-----------------|-------------------------------|--------------------|-------------------|